Active sensing explains cortical variability and improves visual information

Declan P. Rowley[2,*], Ryan A. Ressmeyer[1,*], Tejasvi Kothapalli[3,4,*], Alexander C. Huk[2], Jude F. Mitchell[5], Daniel A. Butts[6], Jacob L. Yates[3,4]

[1]Department of Bioengineering, University of Washington; [2]Fuster Laboratory for Cognitive Neuroscience, Departments of Psychiatry & Biobehavioral Sciences and Ophthalmology, UCLA
[3]Redwood Center for Theoretical Neuroscience, UC Berkeley; [4]Herbert Wertheim School of Optometry and Vision Science, UC Berkeley
[5]Brain and Cognitive Science, University of Rochester; [6]Department of Biology, University of Maryland, College Park; *Equal contribution

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Two debates in visual neuroscience

Fixational eye movements

Feature or bug?

Are these tiny movements motor noise, or part of how we see fine detail[5,6]?

Shared cortical variability

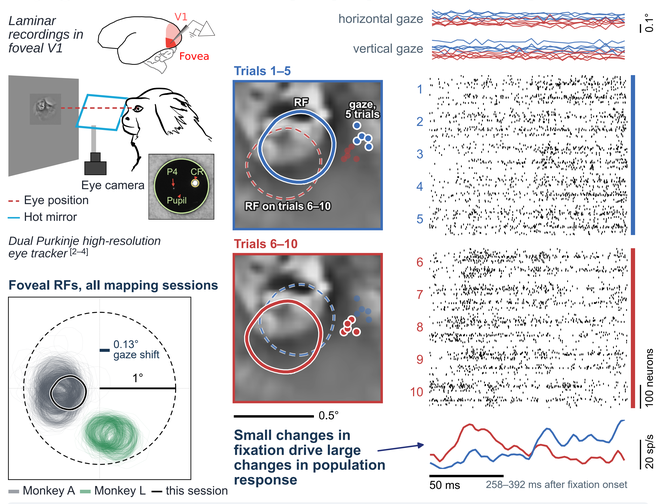
Noise or signal?

The same image evokes different responses. Internal noise, or meaningful sensory input?

How much apparent cortical noise comes from the eye's own movements?

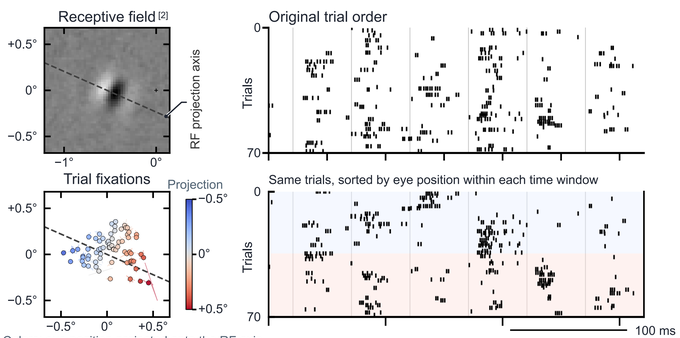
Can that movement-driven activity improve visual information?

1 Tiny gaze shifts reshape population responses



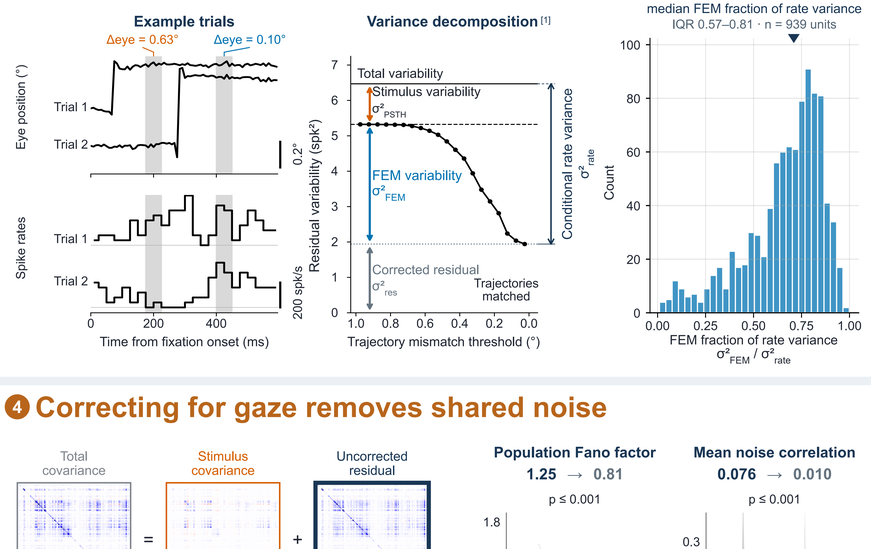
2	1,022	19	20 Hz
awake marmosets	foveal V1 units	recording sessions	repeated natural images

2 One neuron's "noise" tracks the eye



Colors: eye position projected onto the RF axis.

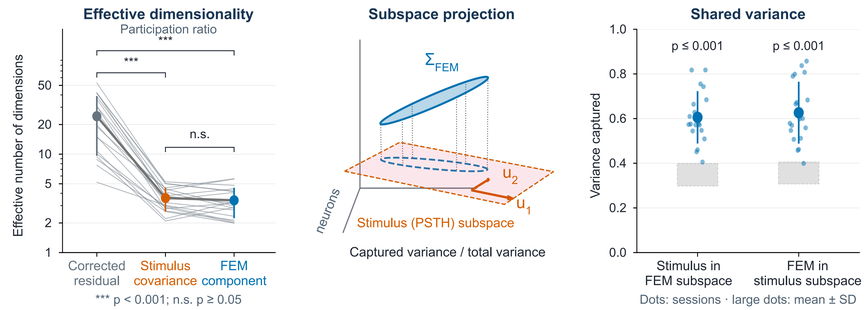
3 Eye movements drive most single-cell rate variance (71%)



4 Correcting for gaze removes shared noise

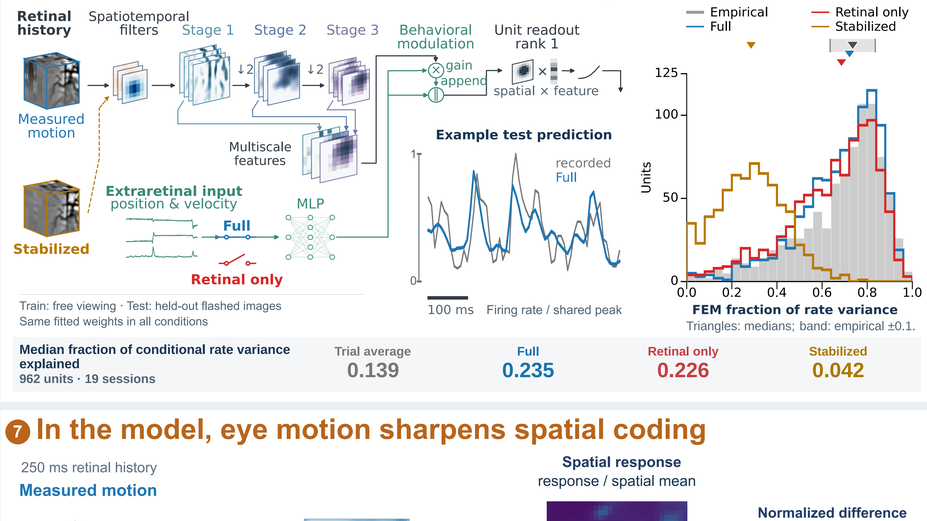
19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

5 Eye-driven variability follows stimulus dimensions



*** p < 0.001; n.s. p ≥ 0.05
25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

6 Gaze-dependent retinal input predicts single trials

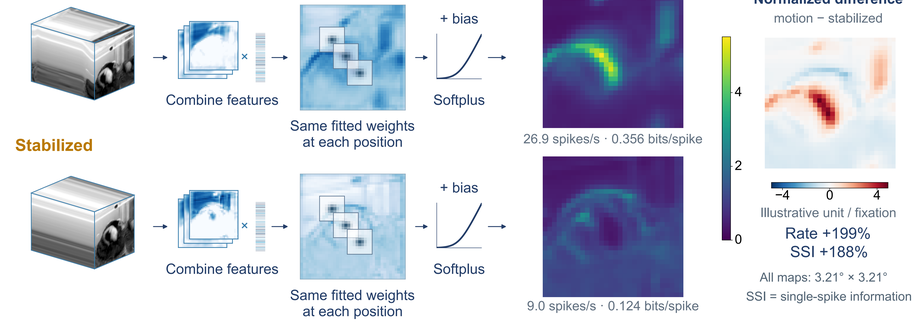


Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions

Triangles: medians; band: empirical ±0.1.

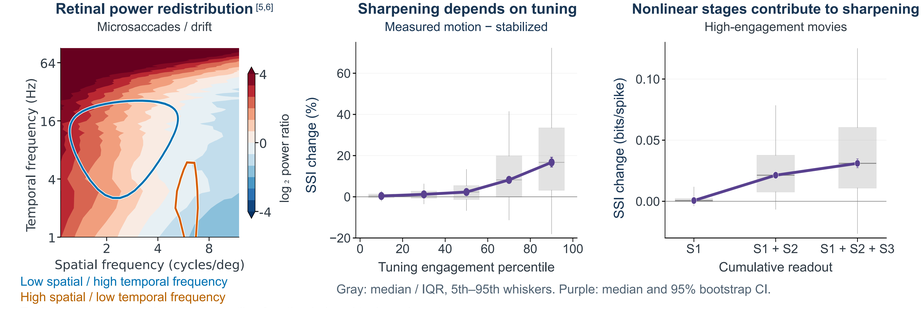
Median fraction of conditional rate variance explained 962 units · 19 sessions	Trial average	Full	Retinal only	Stabilized
	0.139	0.235	0.226	0.042

7 In the model, eye motion sharpens spatial coding



8 Sharpening grows with tuning and model depth

725 units



Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).
Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

Funding
NEI K99-EY037839-01 (D.P.R.); R01-EY030998 (J.F.M.); R01-EY018363 (M.R.).
Fuster Endowment (A.C.H.); NEI K99/R00-EY032179 (J.L.Y.); NIH/NEI T32-EY007043 (T.K.).

Contact
Declan P. Rowley · UCLA
declanrowley@g.ucla.edu

Acknowledgements
Additional preprint coauthors: Dekel Galor · Ruei-Jr Wu · Michele Rucci
We thank Yonatan Abrham (Mitchell Lab) for collecting some of the data used in this report.